Global Food Production Insights

Devangi Dodiya

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1. Introduction

This analysis helps to get insights of food item production across countries and years. There are four dataset as fact_production, dim_country, dim_item, dim_element.. U*sing SQL this project explores global food production data. The goal is to identify trends, inequalities,* *and insights about agricultural production and yield across countries and crops. Total countries worldwide involved are 211, whereas 6 elements to measure total 205 crops and out of them yield and production measured with 2 various units. Total 725837 records from fact_production table contains null or zero for column values.*

2. Methodology

- Tools used: DuckDB
- Data sources (the provided fact_production, dim_country, dim_item, dim_element tables).
- Approach:
  - Write queries to answer predefined challenges.
  - Summarize findings into insights.
  - Present insights in this report and a short video.

3. Analysis & Findings

Challenge 1: Mapping the Giants of Production

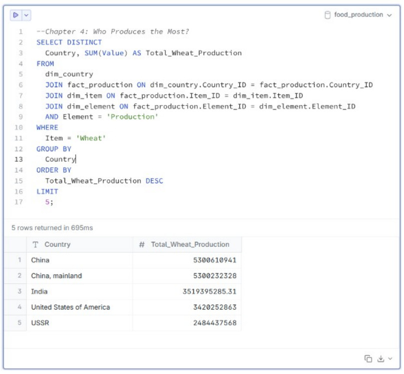
Goal: Which countries lead the charge in growing this food item (Wheat & rice) on a global scale?

SQL Approach: Use all data tables using Join and Item filter of Wheat and Rice

Key Findings:

- "India and China are the largest wheat/rice producers, together accounting for over 40% of global output."
- Production is concentrated in a few countries, suggesting global supply risks if one of them experiences disruptions (e.g., epidemic, war, or trade restrictions).

Output:

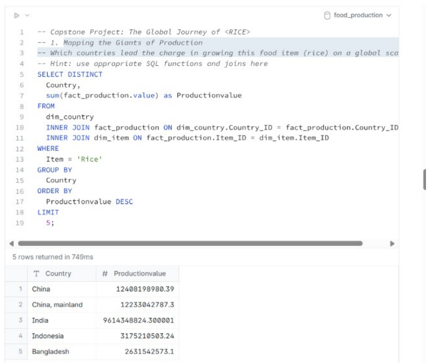


food_production

```
--Chapter 4: Who Produces the Most?
SELECT DISTINCT
  Country, SUM(Value) AS Total_Wheat_Production
FROM
  dim_country
  JOIN fact_production ON dim_country.Country_ID = fact_production.Country_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
  AND Element = 'Production'
WHERE
  Item = 'Wheat'
GROUP BY
  Country
ORDER BY
  Total_Wheat_Production DESC
LIMIT
  5;
```

5 rows returned in 695ms

	Country	Total_Wheat_Production
1	China	5300610941
2	China, mainland	5300232328
3	India	3519395285.31
4	United States of America	3420252863
5	USSR	2484437568



food_production

```
-- Capstone Project: The Global Journey of <RICE>
-- 1. Mapping the Giants of Production
-- Which countries lead the charge in growing this food item (rice) on a global sca
-- Hint: use appropriate SQL functions and joins here
SELECT DISTINCT
  Country,
  sum(fact_production.value) as Productionvalue
FROM
  dim_country
  INNER JOIN fact_production ON dim_country.Country_ID = fact_production.Country_ID
  INNER JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Item = 'Rice'
GROUP BY
  Country
ORDER BY
  Productionvalue DESC
LIMIT
  5;
```

5 rows returned in 749ms

	Country	Productionvalue
1	China	12408198980.39
2	China, mainland	12233042787.3
3	India	9614348824.300001
4	Indonesia	3175210503.24
5	Bangladesh	2631542573.1

Figure 1. SQL Query and Output for Wheat and Rice Producers

Challenge 2: Yield Champions: Efficiency Over Volume

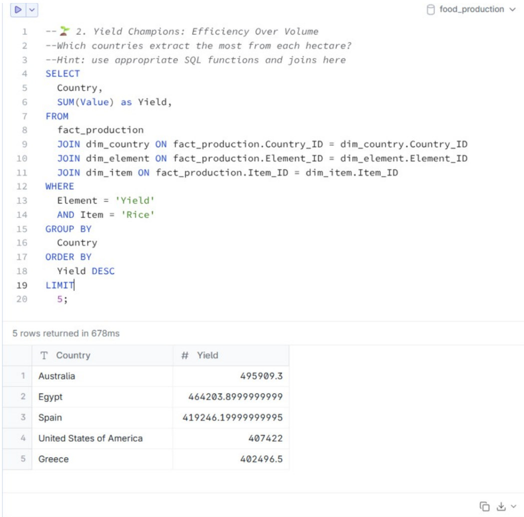
Goal: Which countries extract the most from each hectare?

SQL Approach: Here filtered the dataset to select only records with yield and rice. By joining the fact table with the country, item, and element dimensions, we grouped by country and calculated the total yield values. Finally, we ordered results in descending order and limited the output to the top 5 countries.

Key Findings:

- High yields may indicate better farming technology, irrigation systems, or fertilizer use, whereas lower yields may reflect less mechanization or harsher climates.
- Volume leaders who are biggest total producers, e.g., India, China. Whereas efficiency leaders (countries maximizing land productivity, sometimes smaller nations with advanced farming practices).

Output:


food_production

```sql
-- 🌱 2. Yield Champions: Efficiency Over Volume
--Which countries extract the most from each hectare?
--Hint: use appropriate SQL functions and joins here
SELECT
  Country,
  SUM(Value) as Yield,
FROM
  fact_production
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Element = 'Yield'
  AND Item = 'Rice'
GROUP BY
  Country
ORDER BY
  Yield DESC
LIMIT
  5;
```

5 rows returned in 678ms

	Country	Yield
1	Australia	495909.3
2	Egypt	464203.8999999999
3	Spain	419246.19999999995
4	United States of America	407422
5	Greece	402496.5

Figure 2. SQL Query and Output for Yield Champions: Efficiency Over Volume

Challenge 3: How Has Global Production Shifted Through Time?

Goal: Did wars, climate patterns, or economic booms affect production volumes?

SQL Approach: Using Join the fact table with country, item, and element dimensions and filtered for production and rice. Grouping results by both country and year allowed us to calculate annual production totals. This makes it possible to compare rice production levels across time and detect historical patterns of growth, decline, or volatility.

Key Findings:

- Production trends are not uniform.
- Global rice production shows a steady upward trend across most decades, reflecting population growth and advances in agricultural technology.
- Found that Germany, Denmark and few other countries are not producing rice at all.

Output:

```sql
--3. How Has Global Production Shifted Through Time?
--Did wars, climate patterns, or economic booms affect production volumes?----
--Hint: use appropriate SQL functions and joins here
SELECT
  Country,
  SUM(Value) as Global_Prodcution,
  Year
FROM
  fact_production
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Element = 'Production'
  AND Item = 'Rice'
GROUP BY
  Country,
  Year;
```

9.387 rows returned in 704ms, queued for 680ms

	Country	Global_Prodcution	Year
1	Democratic Republic of the Congo	70800	1961
2	Denmark	NULL	1961
3	Germany	NULL	1961
4	Guinea	210940	1961
5	Haiti	55000	1961
6	Honduras	11928	1961
7	Luxembourg	NULL	1961
8	Niger	9550	1961
9	Papua New Guinea	1600	1961

Figure 3. SQL Query and Output for Global Production Shifted Through Time

Challenge 4: Yield Volatility: Who Plays the Risky Game?

Goal: Which countries show erratic yield swings?

SQL Approach: We filtered the dataset with yield and rice. Using the **STDDEV(Value)** function, we calculated the statistical standard deviation of yields for each country. Higher standard deviation values indicate greater fluctuations, while lower values suggest stability. Results were ordered in descending order to highlight the most volatile producers.

Key Findings:

- Some countries exhibit high yield volatility, meaning their rice output per hectare fluctuates widely from year to year.
- Volatile countries face food security risks such as unpredictable harvests make it harder to plan exports, imports, and domestic consumption.

Output:

food_production

```sql
--🎲 4. Yield Volatility: Who Plays the Risky Game?
--Which countries show erratic yield swings?
--Hint: use appropriate SQL functions and joins here
SELECT
  Country,
  stddev(Value) as Yield_Volatility
FROM
  fact_production
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Element = 'Yield'
  AND Item = 'Rice'
GROUP BY
  Country
ORDER BY
  Yield_Volatility DESC;
```

128 rows returned in 824ms

	Country	Yield_Volatility
1	Tajikistan	2755.7120079858964
2	Puerto Rico	2318.694435780234
3	Syrian Arab Republic	2274.23211420945
4	Uruguay	1912.5765236098646
5	Egypt	1887.7623580284528
6	Uzbekistan	1863.5207429554957
7	Morocco	1808.4589854807043

Figure 4. SQL Query and Output for countries show erratic yield swings

Challenge 5: A Tale of Inequality: Distribution of Global Output

Goal: Is production concentrated or shared globally?

SQL Approach: filtered for production and rice. By grouping results by country, along with calculation of each nation's total rice production. Using a window function **(SUM(SUM(Value)) OVER ())**, we computed the global total and derived each country's percentage share of global rice production. Results were then ranked by production size.

Key Findings:

- The top 5 producers (such as China, India, Indonesia, Bangladesh, and Vietnam) account for a large majority of global rice output.
- Many other countries contribute only small percentages, reflecting strong global dependence on a handful of agricultural giants.
- This concentration creates supply exposer: disruptions in just one major producer (e.g., drought, export bans, or political conflict) can have outsized global impacts.

Output:

food_production

```sql
--⚖️ 5. A Tale of Inequality: Distribution of Global Output
--Is production concentrated or shared globally?
--Hint: use appropriate SQL functions and joins here
SELECT
  Country,
  SUM(Value) AS Country_Production,
  ROUND(SUM(Value) * 100.0 / SUM(SUM(Value)) OVER (), 2) AS Global_Share_Percent
FROM
  fact_production
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Element = 'Production'
  AND Item = 'Rice'
GROUP BY
  Country
ORDER BY
  Country_Production DESC;
```

149 rows returned in 825ms

	Country	Country_Production	Global_Share_Percent
1	China	10414256279.189999	24.2
2	China, mainland	10269786020	23.87
3	India	7005921312	16.28
4	Indonesia	2544142545.54	5.91
5	Bangladesh	1975004918	4.59
6	Viet Nam	1549368029.59	3.6
7	Thailand	1421347270	3.3

Figure 5. SQL Query and Output for countries show Global Output for rice production

Challenge 6: Spotlight: A Country's Journey Over the Years

Goal: How has a country's production changed over time?

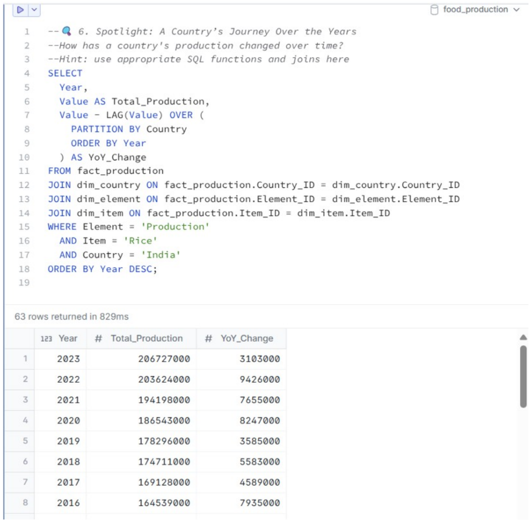
SQL Approach: Here filtering India, production and rice. Using the LAG() window function, we compared each year's production value to the previous year. This allowed us to calculate the Year-over-Year (YoY) Change, highlighting increases or decreases in production across decades.

Key Findings:

- India's rice production shows a long-term upward trend, reflecting agricultural modernization, expanded irrigation, and population-driven demand.
- The YoY metric helps distinguish between steady growth periods and volatile phases in India's agricultural history.

- Year-to-year changes reveal periods of both strong growth and temporary declines: Sharp increases align with agricultural reforms and technological adoption (e.g., Green Revolution effects in the late 20th century). Declines correspond with drought years, monsoon failures, or other climate-related shocks.

Output:



food_production

```sql
-- 🔍 6. Spotlight: A Country's Journey Over the Years
--How has a country's production changed over time?
--Hint: use appropriate SQL functions and joins here
SELECT
  Year,
  Value AS Total_Production,
  Value - LAG(Value) OVER (
    PARTITION BY Country
    ORDER BY Year
  ) AS YoY_Change
FROM fact_production
JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE Element = 'Production'
  AND Item = 'Rice'
  AND Country = 'India'
ORDER BY Year DESC;
```

63 rows returned in 829ms

	Year	Total_Production	YoY_Change
1	2023	206727000	3103000
2	2022	203624000	9426000
3	2021	194198000	7655000
4	2020	186543000	8247000
5	2019	178296000	3585000
6	2018	174711000	5583000
7	2017	169128000	4589000
8	2016	164539000	7935000

Figure 6. SQL Query and Output for India's rice Journey Over the Years

Challenge 7: What Do We Actually Track About This Crop?

Goal: Which kinds of measurements exist for your item?

SQL Approach: Using of joined the fact table with the element and item dimensions and filtered for `Item = 'Rice'`. By selecting `DISTINCT Unit`, this tells that **different measurement units** tracked for rice in the dataset.

Key Findings:

- For rice, the dataset tracks **several different measurement dimensions**, each offering a different perspective on production:
    - ► **Production (tonnes)** – total harvested output.
    - ► **Area harvested (hectares)** – how much land is devoted to rice.
    - ► **Yield (tonnes/hectare)** – efficiency of land use.
- Having multiple units allows us to explore not just *how much* rice is produced, but also *how it is produced* (efficiently or inefficiently, on large or small areas of land).

Output:

food_production

```sql
--📋 7. What Do We Actually Track About This Crop?
--Which kinds of measurements exist for your item?
--Hint: use appropriate SQL functions and joins here
SELECT
  DISTINCT Unit,
  Element,
FROM
  dim_element
  JOIN fact_production ON fact_production.Element_ID = dim_element.Element_ID
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE
  Item = 'Rice'
ORDER BY
  Country,
  Year DESC;
```

3 rows returned in 862ms

	Unit	Element
1	ha	Area harvested
2	kg/ha	Yield
3	t	Production

Figure 7. SQL Query and Output for measurements used to track rice

Challenge 8: Breakout Growth Stories: Year-on-Year Jumps

Goal: Where did production suddenly explode?

SQL Approach: Aggregation of rice production totals by year (SUM(Value)) and used the LAG() window function to calculate the difference between the current year and the previous year. This produced both the absolute YoY change (increase in tonnes) and the percentage YoY change (growth rate). Ordering results by year highlights historical moments of sudden production jumps.

Key Findings:

- Global rice production shows a general upward trend, but certain years stand out with breakout growth.
- These sharp jumps likely to happened due to certain situation of that particular year such as agriculture revolution, economic boom, climate changes.

Output:

food_production

```sql
--📈 8. Breakout Growth Stories: Year-on-Year Jumps
--Where did production suddenly explode?
--Hint: use appropriate SQL functions and joins here
SELECT
    Year,
    SUM(Value) AS Total_Production,
    Total_Production - LAG(Total_Production) OVER (ORDER BY Year) AS YoY_Change,
    ROUND(
        (Total_Production - LAG(Total_Production) OVER (ORDER BY Year))
        * 100.0 / NULLIF(LAG(Total_Production) OVER (ORDER BY Year),0), 2) AS YoY_P
FROM fact_production
JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
WHERE Element = 'Production' AND Item = 'Rice'
GROUP BY Year
ORDER BY Year DESC;
```

63 rows returned in 824ms

	Year	Total_Production	YoY_Change	YoY_Percent_Change
1	2023	1008063704.9499999	9059287.899999976	0.91
2	2022	999004417.05	-6221978.310000062	-0.62
3	2021	1005226395.36	16501478.960000038	1.67
4	2020	988724916.4	21831781.589999676	2.26
5	2019	966893134.8100003	-11065846.129999757	-1.13
6	2018	977958980.94	14630051.169999957	1.52
7	2017	963328929.7700001	13222563.669999957	1.39
8	2016	950106366.1000001	2305492.7999999523	0.24

Figure 8. SQL Query and Output for Production Year-on-Year Jumps

Challenge 9: Share of the Global Pie

Goal: How much does your chosen country contribute to the world's total production each year?

SQL Approach: grouping rice production data by year, summing all countries' values for the global total. Using a conditional aggregation (CASE WHEN Country = 'India'), we separately calculated India's yearly rice production. Dividing India's production by the global total gave India's percentage share of the global pie, tracked over time.

Key Findings:

- India consistently ranks among the largest contributors to global rice production, with a substantial share year after year.
- India's percentage contribution has been relatively stable but shows small upward and downward shifts depending on harvest conditions and global dynamics.
- The data shows that while India is not the sole leader, its contribution is crucial which clearly denotes that fluctuations in India's output strongly impact global availability and prices.
- This reinforces India's role not only as a domestic food provider but also as a key player in international rice trade.

Output:

food_production

```sql
--🌐 9. Share of the Global Pie
--How much does your chosen country contribute to the world's total production each
--Hint: use appropriate SQL functions and joins here
SELECT
    Year,
    SUM(Value) AS Total_Production,
    SUM(
        CASE WHEN Country = 'India' THEN Value ELSE 0 END
    ) AS India_Production,
    ROUND(
        SUM(CASE WHEN Country = 'India' THEN Value ELSE 0 END) * 100.0
        / SUM(Value),
        2
    ) AS India_Percentage
FROM fact_production
JOIN dim_item ON fact_production.Item_ID = dim_item.Item_ID
JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
WHERE Element = 'Production' AND Item = 'Rice'
GROUP BY Year
ORDER BY Year DESC;
```

63 rows returned in 860ms

	123 Year	# Total_Production	# India_Production	# India_Percentage
1	2023	1008063704.9499999	206727000	20.51
2	2022	999004417.05	203624000	20.38
3	2021	1005226395.36	194198000	19.32
4	2020	988724916.4	186543000	18.87
5	2019	966893134.8100003	178296000	18.44
6	2018	977958980.94	174711000	17.86

Figure 9. SQL Query and Output for India's Production in Global Share along with Percentage over Year

Challenge 10: Smoothing the Signal: 3-Year Rolling Averages

Goal: Has production grown steadily or in spurts?

SQL Approach: joining the production fact table with country, item, and element dimensions, filtered for production and rice as item, and calculated a 3-year rolling average using a window function:

Key Findings:

- The rolling averages show steadier trends compared to raw yearly data.

Output:

food_production

```sql
-- 10. Smoothing the Signal: 3-Year Rolling Averages
--Has production grown steadily or in spurts?
--Hint: use appropriate SQL functions and joins here
SELECT
  year,
  Country,
  Value AS Rice_Production,
  ROUND(AVG(Value) OVER (
    ORDER BY
      Year ROWS BETWEEN 2 PRECEDING
      AND CURRENT ROW
  ), 1) AS Rolling_3yr_Avg
FROM
  fact_production
  JOIN dim_item ON dim_item.Item_ID = fact_production.Item_ID
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
WHERE
  Element = 'Production'
  AND Item = 'Rice'
ORDER BY
  Country, year;
```

9.387 rows returned in 848ms

	Year	Country	Rice_Production	Rolling_3yr_Avg
1	1961	Afghanistan	319000	319000
2	1962	Afghanistan	319000	106666.7
3	1963	Afghanistan	319000	107180
4	1964	Afghanistan	380000	3341990
5	1965	Afghanistan	380000	127533.3

Figure 10. SQL Query and Output for India’s Production in Global Share along with Percentage over Year

Challenge 11: Yearly Champions: Who Led Each Year?

Goal: Who were the yearly champions of production?

SQL Approach: Here used the RANK() window function to get rank of each year.

Key Findings:

- China and India consistently appear at the top, showing long-term dominance in rice production.
- The rank data highlights whether global rice production is dominated by a few consistent leaders or shared among shifting champions.
- In some years, other countries briefly rise in rank, often due to favorable climate, technology adoption, or policy shifts.

Output:

food_production

```sql
--11. Yearly Champions: Who Led Each Year?
--Who were the yearly champions of production?
--Hint: use appropriate SQL functions and joins here
SELECT
  year,
  Country,
  Value AS Rice_Production,
  Rank() OVER (
    PARTITION BY
      Year
    ORDER BY
      Rice_Production DESC
  ) as Champions
FROM
  fact_production
  JOIN dim_item ON dim_item.Item_ID = fact_production.Item_ID
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
WHERE
  Element = 'Production'
  AND Item = 'Rice';
```

9.387 rows returned in 794ms

	Year	Country	Rice_Production	Champions
1	1973	China	124584102	1
2	1973	China, mainland	121735008	2
3	1973	India	66077008	3
4	1973	Indonesia	21489500	4
5	1973	Bangladesh	17862704	5

Figure 11. SQL Query and Output for Yearly Champion for Rice Production

Challenge 12: Who Improved the Most Over 5 Years?

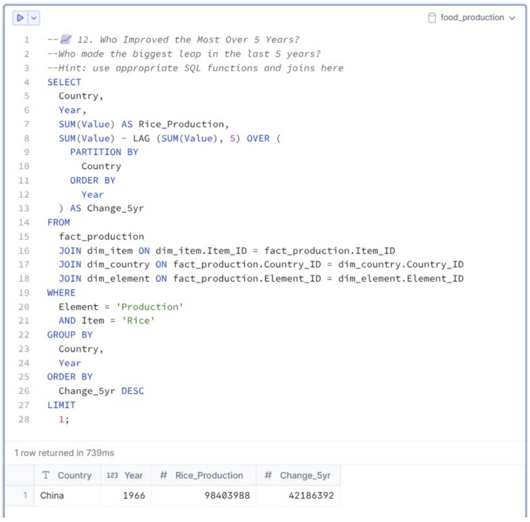
Goal: Who made the biggest leap in the last 5 years?

SQL Approach: Aggregated production (SUM(Value)) per country per year and applied the window function.

Key Findings:

- The query identifies fastest-growing producers, not just the biggest overall producers.
- Countries with the steepest growth often invested in irrigation, new crop varieties, or expanded cultivation.

Output:



food_production

```sql
--📈 12. Who Improved the Most Over 5 Years?
--Who made the biggest leap in the last 5 years?
--Hint: use appropriate SQL functions and joins here
SELECT
  Country,
  Year,
  SUM(Value) AS Rice_Production,
  SUM(Value) - LAG (SUM(Value), 5) OVER (
    PARTITION BY
      Country
    ORDER BY
      Year
  ) AS Change_5yr
FROM
  fact_production
  JOIN dim_item ON dim_item.Item_ID = fact_production.Item_ID
  JOIN dim_country ON fact_production.Country_ID = dim_country.Country_ID
  JOIN dim_element ON fact_production.Element_ID = dim_element.Element_ID
WHERE
  Element = 'Production'
  AND Item = 'Rice'
GROUP BY
  Country,
  Year
ORDER BY
  Change_5yr DESC
LIMIT
  1;
```

1 row returned in 739ms

	Country	Year	Rice_Production	Change_5yr
1	China	1966	98403988	42186392

Figure 12. SQL Query and Output for China as Who Biggest Leap in the Last 5 Years

4. Discussion

- Production is highly concentrated, meaning global supply depends on a few countries such as for rice and wheat china and India are on top. Countries like India and China are not just domestic producers but global influencers, shaping trade flows and food security.
- Using window functions, rolling averages, and aggregate measures gave deeper insight than raw totals, making hidden patterns visible.

5. Conclusion

This project demonstrated how SQL can uncover powerful insights from agricultural production data, moving from simple queries to advanced analytics with joins, aggregates, and window functions. By examining rice production across countries and years, several key themes emerged:

Global Concentration as a small group of countries, led by China and India, dominate global rice production.

rice production is both concentrated and dynamic. Established leaders ensure global stability, but emerging producers and shifting yields reveal a constantly evolving agricultural landscape. For policymakers, traders, and researchers, these insights are vital for anticipating risks and opportunities in the global food system.